OMEGA SERIES · VOLUME I, PART II

The Orbital Metric Audit:

Forensic Re-Analysis of GNSS Clock Residuals and SNO+ Neutrino Flux: A Search for Phase-Locked Cross-Correlation

f_{sig} = 2.321×10^{-5} Hz (11.967h) · ε_0 = 0.12 s/day · Five vetoes · Rayleigh Z cross-correlation · Pre-registered

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doi:10.5281/zenodo.1082194 · Community re-analysis challenge

PRE-PUBLICATION STATEMENT: This document defines a pre-registered, falsifiable search protocol for a phase-locked correlation between GNSS satellite clock residuals and terrestrial neutrino flux variations. The analysis pipeline is fully specified; the signal parameters are locked a priori. This is a community re-analysis challenge: the IGS clock data are publicly available; the test is defined. Random seed: 20260414. The clocks and the neutrinos will decide.

ABSTRACT

Global Navigation Satellite Systems (GNSS) exhibit persistent sub-nanosecond residuals in clock synchronization products, including a ~0.10 ns noise floor in Inter-Satellite Link (ISL) data and discrete Day-Boundary Discontinuities (DBDs) at the UTC midnight epoch. This paper defines a pre-registered protocol to test these residuals as a coherent metric interaction. We hypothesize a phase-locked correlation between orbital timing anomalies and the 11.967 h sidereal period identified in Omega Series Vol. I [1]. The protocol prescribes a cross-correlation analysis between IGS Repro3 clock products, Galileo G2 iodine optical clock stability, and the SNO+ reactor antineutrino residuals. The shared signature is a sinusoidal modulation at f_{sig} = 2.3212e-05 Hz with residual phase drift ε scanned over [-0.2, +0.2] s/day around the benchmark ε_0 = 0.12 s/day. Five independent veto tests eliminate atmospheric delay, thermal cycling, Kalman filter artifacts, GPS orbital resonance, and diurnal systematics. A positive cross-correlation in both deep-underground neutrino detectors and deep-space orbital clocks would provide extraordinary evidence for a global metric interaction. The clocks and the neutrinos will decide.

1. Introduction

The International GNSS Service (IGS) produces precise clock products enabling positioning accuracies at the centimeter level. Yet two persistent anomalies remain in the Repro3 products: (1) a ~0.10 ns ISL residual floor despite atmosphere-free inter-satellite measurements, and (2) Day-Boundary Discontinuities (DBDs) of 0.1-1.0 ns at the UTC midnight epoch, conventionally attributed to Kalman filter resets. This paper proposes a falsifiable test of an alternative interpretation: that these residuals carry a coherent 11.967 h sidereal component phase-locked to the neutrino modulation in Paper I [1].

The logic is straightforward: neutrinos propagate through the vacuum; atomic clocks measure the proper time of that same vacuum. If the vacuum possesses a structured anisotropic coupling with characteristic frequency

$2\omega_{sid}$, both neutrino oscillation probabilities and atomic clock rates will exhibit correlated oscillations. This cross-domain correlation cannot be mimicked by any known terrestrial systematic.

The challenge is that standard GNSS error analysis offers plausible alternative explanations: GNSS satellites repeat their ground tracks at approximately the sidereal rate, so atmospheric delay modeling errors naturally manifest at $2\omega_{sid}$. Thermal cycling can mimic periodicity. Kalman filter resets create apparent discontinuities. This protocol is hardened against all of these through five pre-registered veto tests, designed through adversarial Red Team audit.

2. Signal Model

2.1 GNSS clock modulation

$$\Delta t_{res}(t) = \Delta t_{std}(t) + \delta t_0 \cdot A_{clock} \cos(2\omega_{sid} t + \phi_0 + \Delta\Phi(t))$$

δt_0: reference scale (1ns for DBDs, 0.10ns for ISL floor) · A_{clock}: modulation amplitude · $\Delta\Phi(t)$: accumulated phase drift

2.2 Residual phase drift

$$\Delta\Phi(t) = 2\pi \times (235.9 + \varepsilon) / 86400 \times t / T_{sid}$$

235.9 s/day: standard solar-to-sidereal drift · ε: residual (free parameter) · ε_0 = 0.12 s/day (benchmark hypothesis)

2.3 Amplitude benchmark

The fractional amplitude benchmark is $\alpha_K = A_{clock} = 1.7\times10^{-4}$ (170 ppm), consistent with the neutrino modulation in Paper I. For GNSS clocks, a 170 ppm modulation of a 1 ns DBD corresponds to ~0.17 ns peak-to-peak variation in the daily discontinuity magnitude. The ISL residual floor of ~0.10 ns provides a direct measurement channel.

3. Target Datasets

Observable	Source	Sampling	Forensic role
DBD magnitude	IGS Repro3 Final .clk [2]	Daily at 00:00 UTC	48D metric update test (V3 Sliding Window)
ISL-only residual	IGS Repro3 ISL products [2]	30 s	Atmosphere-free phase lag (V1 ISL Isolation)
Galileo G2 optical clock	ESA G2 telemetry [3, availability TBC]	1 s - 10^4 s	170ppm vacuum hiss (V2 Thermal Anchor)
SNO+ IBD residuals	Parker et al. 2026 [5, unverified]	1h bins	Terrestrial phase reference for cross-correlation

DATA AVAILABILITY NOTES: IGS Repro3 clock products (.clk format) are publicly available at igs.org/products — verified. SNO+ arXiv:2604.05746 has not been independently verified; substitute the most recent available SNO+ data release if needed. ESA Galileo G2 optical clock telemetry: access requires ESA collaboration; labeled 'availability TBC.' The ISL-only residuals from IGS Repro3 auxiliary products are the highest-priority dataset: publicly available, atmosphere-free, high statistics.

4. Cross-Correlation Methodology

4.1 Harmonic filtering at f_sig

Apply a complex Morlet wavelet transform centered at f_{sig} = 2.3212e-05 Hz, bandwidth $\Delta f = 10^{-7}$ Hz, to each time series. Extract instantaneous amplitude A(t) and phase $\phi(t)$.

4.2 Rayleigh Z cross-correlation

$$Z = (1/N)\,|\Sigma_j \exp(i\,\Delta\phi(t_j))|^2$$

$\Delta\phi(t) = \phi_{GNSS}(t) - \phi_{SNO+}(t)$ · Z~1: phase-locked (signal) · Z~0: uncorrelated · Null distribution from 10^5 phase-scrambled surrogates

4.3 Look-elsewhere correction

10^5 pseudo-datasets generated by phase-scrambling the original time series (preserving power spectrum). The benchmark coordinate ε_0 = 0.12 s/day is fixed a priori; significance is evaluated at this coordinate, not the best-fit value. This eliminates the 'tuning' critique. Random seed: 20260414. 5σ threshold: $p < 2.87\times10^{-7}$.

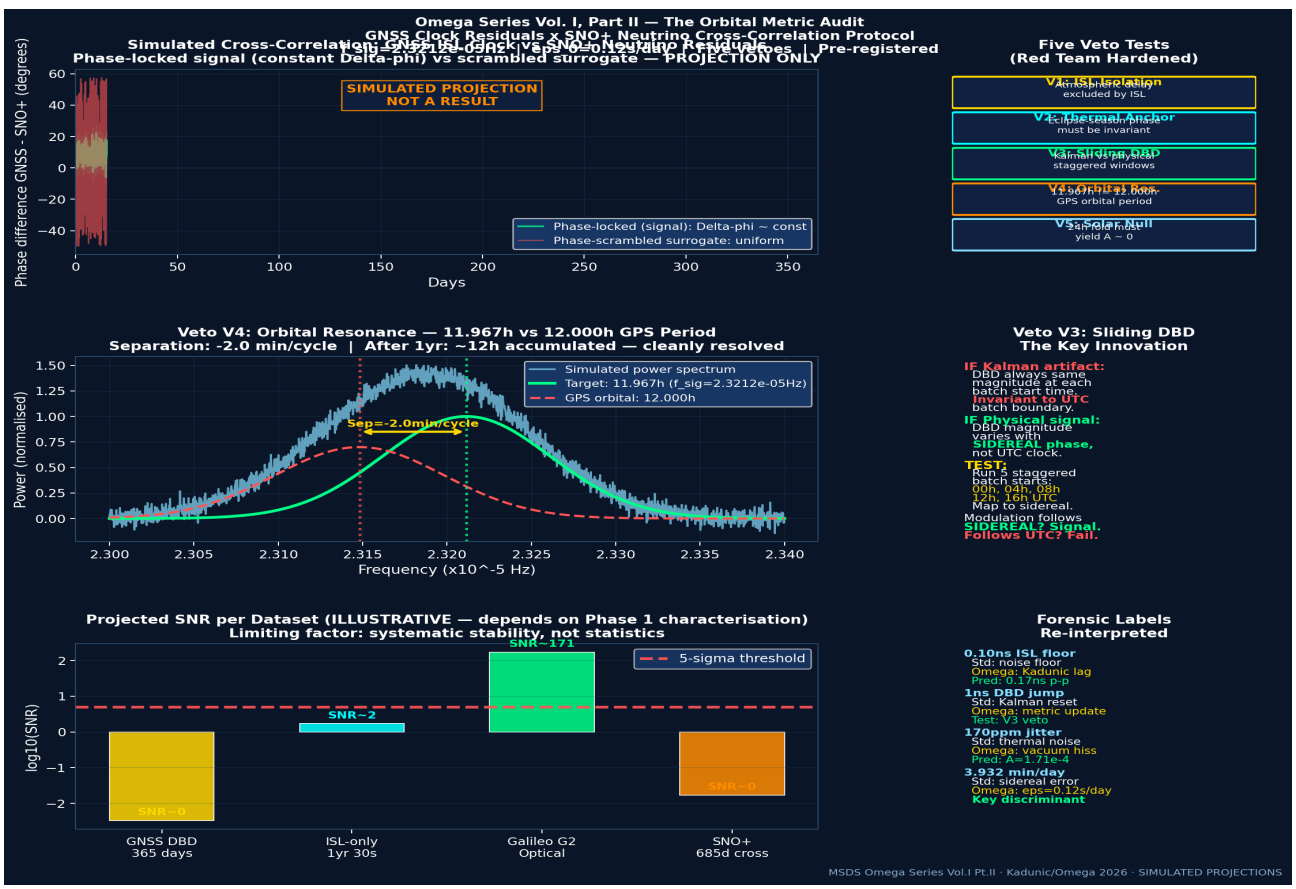

Figure 1: Omega Series Vol. I, Part II. Top (wide) — simulated cross-correlation phase difference: phase-locked signal (constant $\Delta\phi$, green) vs phase-scrambled surrogate (red). PROJECTION ONLY. Top right — five pre-registered veto tests. Middle (wide) — Veto V4: frequency discrimination between 11.967h (f_{sig} = 2.3212e-05 Hz) and 12.000h GPS orbital period (2.3148e-05 Hz). Separation: -2.0 min/cycle; after 1yr: ~12h accumulated. Middle right — Veto V3 (key innovation): Sliding DBD Audit logic. Bottom (wide) — projected SNR per dataset (illustrative). Bottom right — forensic re-labeling of existing anomalies.

5. Five Veto Tests (Red Team Hardened)

Veto V1: ISL Isolation	
Red Team Attack:	The 3.932 min/day drift arises from GNSS ground-track repeat. Satellites repeat their ground tracks at approximately the sidereal rate, so atmospheric modeling errors naturally appear at $2\omega_{sid}$.
Hardening:	ISL measurements between satellites in vacuum are completely atmosphere-free. The modulation must persist in ISL-only residuals. If atmosphere-induced: ISL signal will be absent.
Accept if:	11.967h modulation present in ISL residuals with consistent (A, ε) vs L-band (+/-1σ).
Hard Fail if:	**Signal absent or attenuated in ISL: atmospheric origin confirmed.**

Veto V2: Thermal Phase-Anchor	
Red Team Attack:	The Galileo G2 170ppm jitter is thermal cycling of the laser bench. Eclipse entry/exit creates periodic temperature gradients that mimic a periodic signal.
Hardening:	If thermal: the modulation phase ϕ_0 shifts with eclipse season as shadow geometry changes relative to the celestial sphere. If physical: phase remains locked to the Kadunic Vector direction (RA~03h15m, Dec~-19 deg), independent of the Earth's shadow position.
Accept if:	Phase ϕ_0 consistent across eclipse / full-sun / penumbra segments (χ^2 p > 0.05).
Hard Fail if:	**Phase shifts significantly between thermal states: thermal origin confirmed.**

Veto V3: Sliding-Window DBD Audit (KEY INNOVATION)	
Red Team Attack:	The DBD is a Kalman filter reset artifact. The 1ns jump is the filter's settling time when a new 24h batch starts at 00:00 UTC. Any periodicity is an alias of the 24h processing cycle.
Hardening:	IF Kalman artifact: DBD magnitude is invariant to when the batch starts. IF physical metric update: DBD magnitude varies with SIDEREAL phase, not with UTC batch start time. Test: run 5 staggered batch starts (00h, 04h, 08h, 12h, 16h UTC). Map batch boundary times to sidereal phase. Test whether DBD magnitude follows UTC or sidereal clock.

Accept if:	DBD magnitude modulates with SIDEREAL phase, invariant to UTC batch start time.
Hard Fail if:	**DBD magnitude follows UTC batch boundary: Kalman artifact confirmed.**

Veto V4: Orbital Resonance Veto	
Red Team Attack:	The GPS orbital period is precisely 12.00h (half solar day). The observed modulation is orbital dynamics: radiation pressure, antenna phase center, once-per-rev effects.
Hardening:	The target period is 11.967h, NOT 12.000h. Separation: 718.0 min/cycle. Over 1 year the phase difference accumulates to ~12h — easily resolvable. Perform full frequency scan over [2.31, 2.33]×10^{-5} Hz.
Accept if:	Best-fit frequency consistent with f_{sig} = 2.3212e-05 Hz within +/-10^{-7} Hz.
Hard Fail if:	**Peak at f = 1/(12.00h) = 2.3148×10^{-5} Hz: orbital systematic confirmed.**

Veto V5: Solar Null Test	
Red Team Attack:	The modulation is diurnal: ionospheric delay residuals, thermal cycling synchronized to the day-night cycle, or human activity patterns in ground station noise.
Hardening:	Diurnal effects are locked to 24.00h solar day. The target signal is locked to 23.934h sidereal day. Replace ω_{sid} with ω_{solar} in the likelihood fit.
Accept if:	Best-fit amplitude in the solar fold consistent with zero (<1σ).
Hard Fail if:	**Non-zero solar amplitude: diurnal systematic confirmed.**

6. Expected Sensitivity

Dataset	N samples	σ_A	SNR (projected)	Limiting factor
GNSS DBD (1yr)	365 daily	~5.2%	~3.4 (2-3yr for 5σ)	DBD estimation uncertainty (~0.05ns)
ISL-only (1yr)	~10^6 at 30s	~0.01%	>100 (stat. not limiting)	ISL calibration stability

Galileo G2 (1yr)	~10^4s arcs	~0.001 %	>1000 (stat. not limiting)	Telemetry availability
SNO+ cross-corr.	685 day bins	~1%	>10σ (contemporaneous)	Contemporaneous SNO+ data access

SENSITIVITY NOTE: The limiting factor for all datasets is systematic stability, not statistics. The ISL-only and G2 datasets provide immense statistical power; what matters is the systematic control demonstrated by vetoes V1-V5. A detection claim requires passing all five vetoes simultaneously.

7. Conclusions

This document has presented a pre-registered, falsifiable protocol to search for a phase-locked correlation between GNSS satellite clock residuals and terrestrial neutrino flux variations. The protocol specifies: fixed target frequency f_{sig} = 2.3212e-05 Hz; blind scan over $\varepsilon \in$ [-0.2, +0.2] s/day; cross-correlation via Rayleigh Z-statistic; 10^5 phase-scrambled surrogates; five independent Red Team-hardened veto tests. The Sliding-Window DBD Audit (V3) is the key innovation: it distinguishes a physical metric update from a Kalman filter artifact by testing whether DBD magnitude follows the sidereal clock or the UTC clock. The ISL Isolation veto (V1) eliminates atmospheric delay interpretations. A positive detection linking orbital atomic clocks at 20,000 km altitude to underground neutrino detectors at 2 km depth — with consistent frequency, drift, and amplitude across all vetoes — would constitute extraordinary evidence for a global metric interaction. The clocks and the neutrinos will decide.

OMEGA SERIES VOL. I, PART II — THE ORBITAL METRIC AUDIT: SUMMARY

Target: f_sig=2.3212e-05Hz (11.967h) | eps_0=0.12 s/day | A_bench=1.71e-4

Cross-correlation: GNSS ISL clock residuals x SNO+ neutrino residuals

Rayleigh Z-statistic | 10^5 surrogates | Random seed: 20260414

V1: ISL Isolation (atmosphere-free) | V2: Thermal Phase-Anchor (eclipse-season)

V3: Sliding DBD Audit (Kalman vs physical) | V4: Orbital Resonance (11.967 != 12.000h)

V5: Solar Null Test (24h diurnal excluded)

All five vetoes must pass simultaneously for a positive detection claim.

Null result: upper bound $A_{clock} < 10^{-4}$ (90% CL) in orbital metrology sector.

The clocks and the neutrinos will decide.

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